Flash Upgrade Specification For AVR/SoundBar SDK

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ANALOG DEVICES

Revision History:

S. No	Version No.	Author	Section(s) Affected	Change(s) Made	Approved Date	Approved By
1	1.0	Vaishnavi H	All	Initial	14th Aug 2024	Ramamoorthy
2	1.1	Vaishnavi H	Procedure for Wav file generation	3.1	18th Sep 2024	Ramamoorthy

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Preface

Thank you for licensing the AVR/Soundbar 21593SDK05 from Analog Devices, Inc. With a rich, powerful instruction set, floating point precision and high-speed execution, it makes SHARC processors a popular choice for designers of signal processing systems.

The AVR/SoundBar SDK is designed to be used in conjunction with the Cross Core® Embedded Studio (CCES) development tools environment and reference platform with an ADSP-21593 SHARC processor. The CCES development environment aids advanced application code development and debug, such as:

- Create, compile, assemble, and link application programs written in C++, C, and assembly
- Load, run, step, halt, and set breakpoints in application programs
- Read and write data and program memory
- Read and write core and peripheral registers

Access to the processor from a personal computer (PC) is achieved through an external JTAG emulator.

Analog Devices JTAG emulators offer faster communication between the host PC and target hardware. Analog Devices carries a wide range of in-circuit emulation products. To learn more about Analog Devices emulators and processor development tools, visit the following site.

http://www.analog.com/en/design-center/landing-pages/001/sharc-processors-software-and-tools.html

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One Technology Way
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Norwood, MA 02062-9106
USA
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Product Support

For any queries suggestions and feedback, contact us at:

Email: `support_hpma@analog.com`

Supported Processors

This evaluation system supports Analog Devices ADSP-21593 processor

Product Information

Product information can be obtained from the Analog Devices Web site and Cross Core Embedded Studio online Help system.

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1. INTRODUCTION

1.1 PURPOSE

Audio applications such as AVRs and SoundBars use flash devices where the audio processing modules' binary image is stored. This binary image is loaded into the DSP during the booting process and executes from DSP in run time. These binary images may have to be updated when the audio processing module software adds a new feature update, bug fix, code updates from IP holders etc., This document explains the method of updating the serial flash memory connected to ADSP-21593 processor via I2S mode with the I2S stream provided by a SOC in a typical sound bar system.

1.2 ACRONYMS, ABBREVIATIONS & KEY WORDS

- *I2S* – Inter IC Sound - Standard protocol for transmitting two channel PCM over a 3-wire serial bus.
- *DSP* –Digital Signal Processor. Throughout this document DSP refers to ADSP-21593.
- *HOST* – Micro controller that acts as master and controls DSP.
- *SOC* – System on Chip.
- *WAV* – Audio file format.
- *CCES* – Cross Core Embedded Studio.

2. Hardware Connectivity

2.1 Block diagram for flash upgrade in sound bar system

ADSP-21593 processor is connected to BD-Player via I2S. The BD-Player will send the data in PCM format to DSP. Host Micro acts as the master in the system and controls the DSP. DSP is connected to Host Micro via SPI0. When DSP has to interrupt Host Micro, it raises FLAG0 connected to IRQ of Host. DSP is connected to Serial Flash via SPI2.

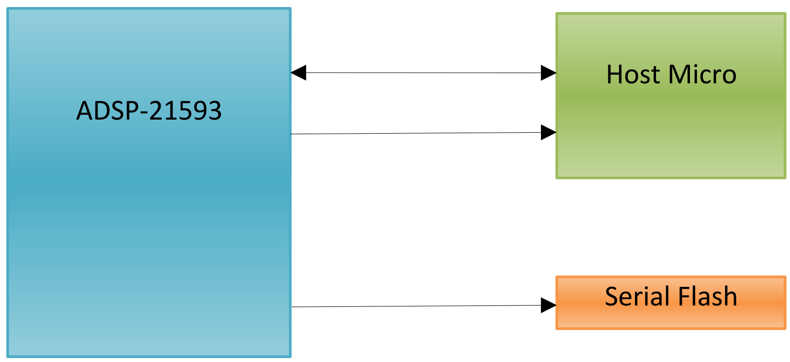

Fig1. Hardware connectivity for flash upgrade on AVR/SoundBar system

2.2 Partitioning of Serial Flash & Contents description

The serial flash connected to DSP is partitioned into two parts. The first part on the top contains the custom boot loader patch and the flash writer software. The second part of the serial flash is allotted for the application code where the binary image of audio application software is stored.

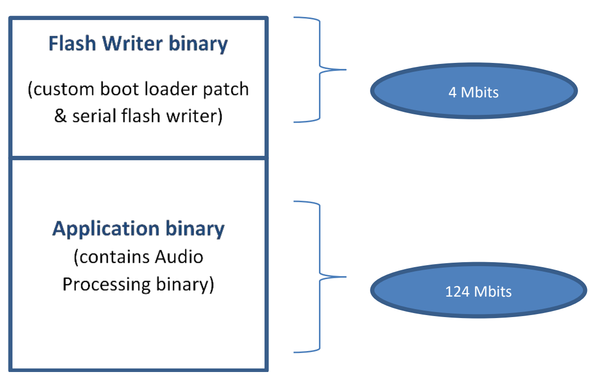

Fig2. Serial Flash Partitioning

Flash writer binary consists of two functions integrated: Custom boot loader patch and the serial flash writer.

Custom boot loader patch: The default boot loader will be customized to process commands from Host micro controller. Depending on the command from Host, the customized boot loader patch will load the serial flash writer code or the application binary.

Serial flash writer: The flash writer software that receives data in the DSP and writes to Serial Flash, reads back and verifies and informs the status of flash writing to Host.

1.2.2. Application binary:

This binary consists of the Audio Decoding modules, audio post processing functions and all the software that is needed to process the audio and playback when an input stream is fed to the product.

2.3 COMMANDS & STATUS INFORMATION:

Flash:

S.No.	Command & Parameter	Command Name	Description
1	Parameter1: 0x0	Sector Number	Sector No to Flash the Application
2	Command: 0xA	FLASH WRITER MODE	Flash the Application to Flash Memory

Download:

S.No.	Command & Parameter	Command Name	Description
1	Parameter1: 0x4	Sector Number	Sector No to download the Application
2	Command: 0xB	APPLICATION MODE	Download the Application binary to DSP

S.No.	Status #	Bits	Description
1	0x23	0:1	00 – Reserved 01 – CRC check passed 10 – CRC check failed 11 – Reserved
		2:3	00 – Reserved 01 – Flash Update completed Successfully 10 – Flash Update Failed 11 – Reserved

2.4 Application Binary format:

The loader file is generated from the CCES tool for the application code. To the loader, 128 bit start sync word, 128 bit end sync word, crc word and zeroes are packed as given in Fig. 3 below.

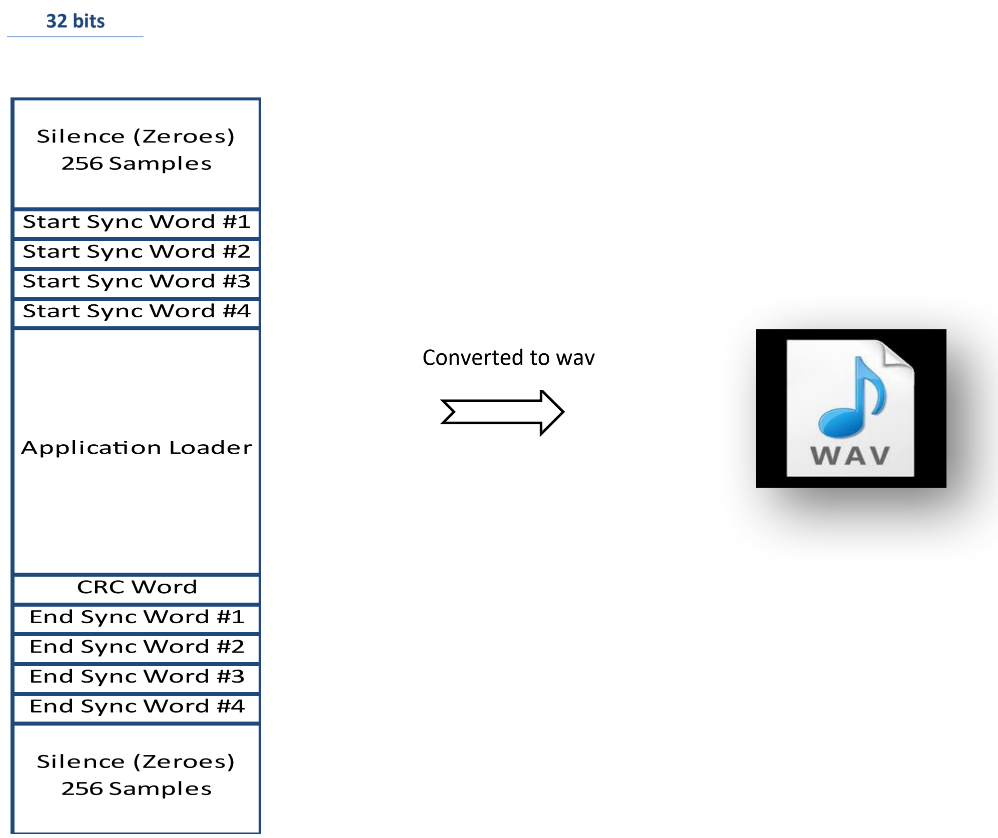

Fig. 3 Application binary format

Wav file format details:

S.No.	*Parameter*	*Value*
1	Sampling Frequency	48 kHz
2	Bits/sample	16
3	Number of channels	2
4	Bit rate	6144Mb/sec

Flash Upgrade protocol:

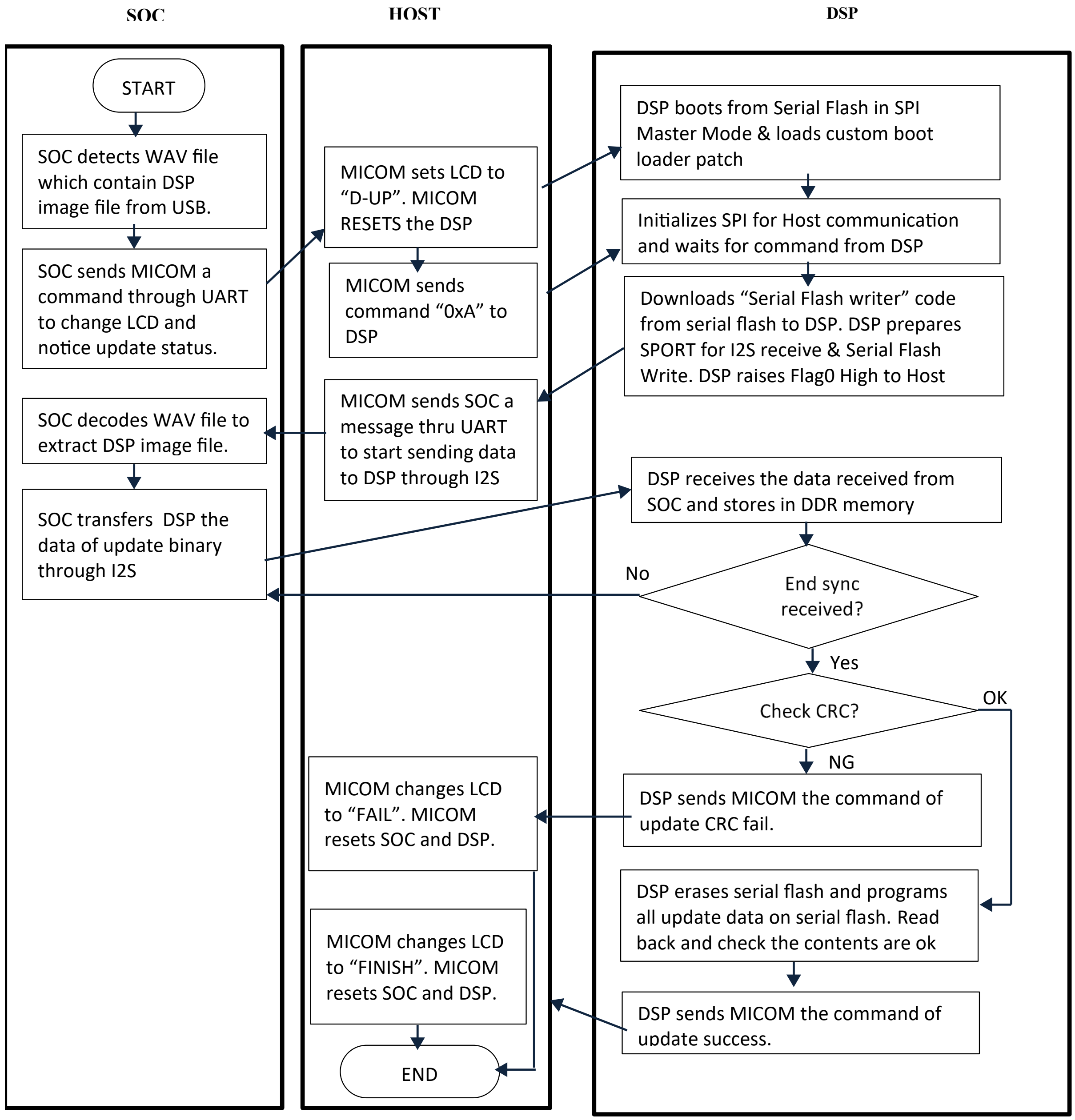

Fig4. Flash Upgrade Protocol

3. Utility for generating the wav file

The loader file for the application code will be generated from CCES. This loader file will be fed to the utility as an input. The utility appends pre-silence, start sync word, crc word, end sync word and post-silence and generates a wav file that can be stored on a USB device and played back by the SOC. From SOC the wav file data are transferred to DSP via I2S as depicted in Fig.4.

3.1 Procedure for Wav file generation

STEP 1: Open the flash upgrade utility as per the below image.

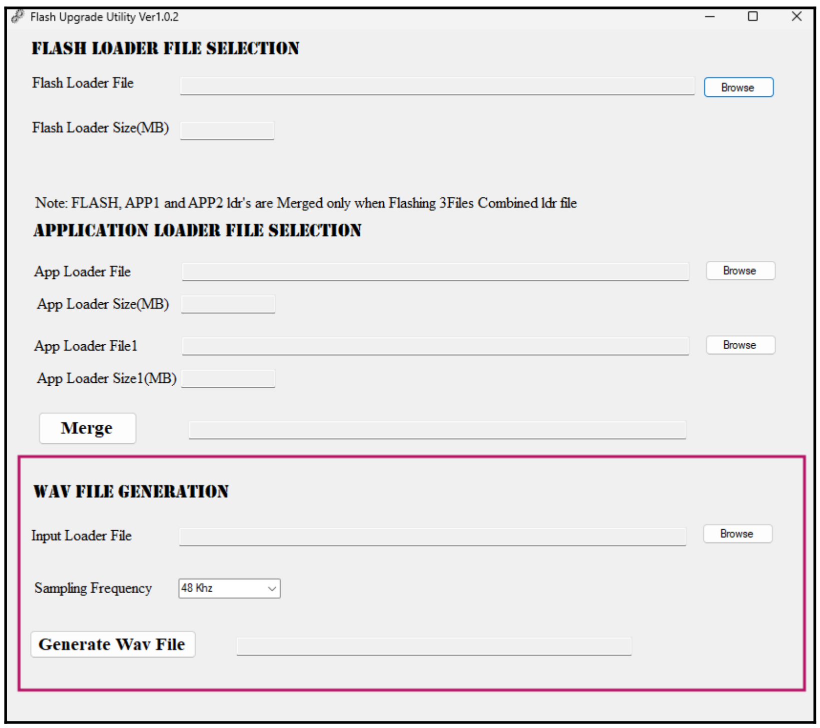

Fig5. Flash Upgrade Utility

STEP 2: In Fig 5, WAV FILE GENERATION -> Input Loader file -> click the Browse -> Select the loader (.ldr) file (from flash writer) -> set Sampling Frequency to 48 Khz -> click the Generate wav file -> The .wav file will be generated in loader (.ldr) file path.

STEP 3: Replace flash writer loader (.ldr) file with the flash upgrade loader (.ldr). The Loader file name should be maintained as like flash writer.

STEP 4: Build the flash writer and flash the code to the board.

STEP 5: Reset the board.

STEP 6: Send the flashing command and then run the generated wav file through the foobar on loop. The LED D4 glows initially before flashing. On flashing completion, LED D4 is turned OFF and LED D3 will glow.

STEP 7: Reset the board.

STEP 8: Send the Booting TTL, LED D3 will blink and then play the streams.

4. SYNCHRONIZATION & ERROR DETECTION

A unique synchronization word of 128 bits, say 'start sync' is added prior to the application loader content. This sync word marks the beginning of the loader data that should be written to the serial flash. Similarly, towards the end of the stream, just after the application loader content, an end sync of 128bit length is added. The 'end sync' marks the end of data write to serial flash.

The data integrity is checked using the embedded CRC. The CRC is calculated for the application loader content only i.e., excluding silence, sync words and crc word. The embedded CRC word is compared against the calculated CRC and if found to be matching, DSP continues with serial flash writing. If there is mismatch in the CRC embedded and CRC calculated, DSP notifies Host of the CRC check failure and refrains from writing to the serial flash.

5. Q & A

1. If GL-XP binary in DSP serial flash is broken, how does GL-XP work?
2. If GL-XP can do nothing in the situation binary broken, please prepare the emergency mode (recovery mode).

Preventive measure: The "Flash writer binary" of Serial flash should be locked after factory flashing. DSP flash writer should have a check to avoid writing to serial flash in the address range of 'flash writer binary' section of serial flash.

Alternative way: There can be a Host connection to serial flash, whereby Host can write the "flash writer binary' to the serial flash. And then the flash writer binary can take care of writing the application binary to the serial flash.